

Lower bounds on the mix norm of passive scalars advected by incompressible energy and enstrophy-constrained flows and Anomalous dissipation

ADELOWO ADEBANJI O

MIXING

What is Mixing?

- Mixing is the reduction of in-homogeneity in order to achieve a homogeneous state, generally this means achieving a uniform distribution of the mixtures.
- The in-homogeneity can be temperature or concentration etc.

Why Mixing?

- Mixing is important in many aspects of science and engineering.
- The understanding of the underlying kinematics of mixing is important because it helps in building and optimization of mixing devices.

MEASURES OF MIXEDNESS

The quantities to describe the degree of mixing are:

- intensity of segregation = variation in concentration
 - Homogenization is mixing by the reduction of intensity of segregation.
 - A mixed state is a state where variance is zero.
- scale of segregation = distribution of length scales.
 - Filamentation is mixing by the reduction of scale of segregation.

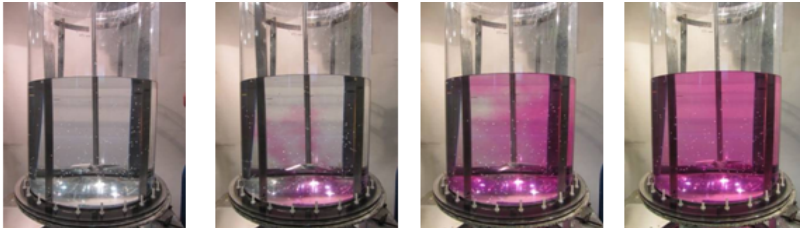


Figure: Mixing by homogenization: Taken from “What does well mixed mean?” by Prof. Suzanne Kresta, University of Alberta, San Francisco, 2013

Illustration I

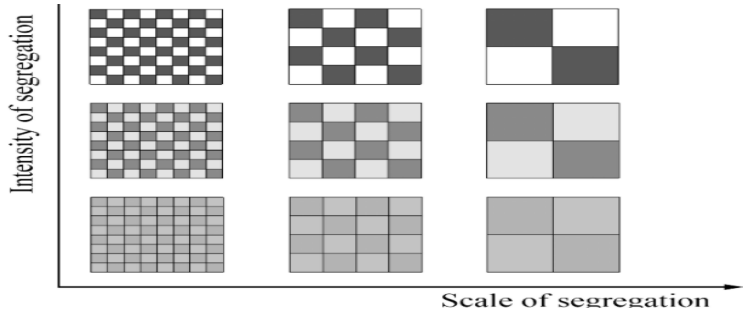


Figure: Taken from “What does well mixed mean?” by Prof. Suzanne Kresta, University of Alberta, San Francisco, 2013

Variance only measures the concentration variance or the intensity of segregation - not the arrangement of the areas. This requires a second measure: the scale of segregation.

Variance as a measure of mixing



$$\text{var}\theta = ||\theta||_{L^2}^2 - \langle\theta\rangle^2 \quad (0.1)$$

where $||\theta||^2 = \frac{1}{|\Omega|} \int_{\Omega} \theta^2 d\Omega$, $\langle\theta\rangle = \frac{1}{|\Omega|} \int_{\Omega} \theta d\Omega$

- Advection-diffusion equation

$$\frac{\partial\theta}{\partial t} + u.\nabla\theta = k\Delta\theta \quad (0.2)$$

- Multiply (0.2) by θ and then integrate by parts and using the incompressibility condition, we have

$$\frac{d}{dt}\langle\theta\rangle = 0 , \quad \frac{d}{dt}||\theta||_{L^2}^2 = -2k||\nabla\theta||_{L^2}^2$$

$\implies$

$$\frac{d}{dt}\text{var}\theta = -2k||\nabla\theta||_{L^2}^2 \quad (0.3)$$

- R.H.S is negative-definite unless θ is constant.
- Rate of mixing: $2k||\nabla\theta||_{L^2}^2$

Variance's Limitation

Observation I

- Although u does not appear in (0.3); its role is to increase the gradients of concentration. Then diffusion acts stronger on the small scales.

What happens in the absence of diffusion?

- $k = 0$: variance is conserved and does not change with time.
- it is sensible to consider

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = 0 \quad (0.4)$$

In this case, we can not use the variance, we will look for another measure of mixing!

Better Measure: H^{-a} norm

Mathew, Mezic and Petzold introduced the mix-norm, which for mean-zero functions is equivalent to $\|\theta(., t)\|_{H^{-\frac{1}{2}}}$. Charles Doering and Lin. Thiffeault [6] generalized the mix-norm to $\|\theta(., t)\|_{H^{-a}}$, $a > 0$.

Definition: Negative Sobolev Norm

The negative Sobolev norm of θ is given as

$$\|\theta(., t)\|_{H^{-a}}^2 = \|\nabla|^{-a}\theta(., t)\|_{L^2}^2 = \sum_{\mathbf{k} \neq 0} k^{-2a} |\hat{\theta}_{\mathbf{k}}(t)|^2 \quad (a > 0)$$

where $k = |\mathbf{k}|$ and $\hat{\theta}_{\mathbf{k}}(t) = \frac{1}{L^{\frac{d}{2}}} \int_{[0,L]^d} e^{-i\mathbf{k}\cdot\mathbf{x}} \theta(\mathbf{x}, t) d\mathbf{x}$

Advantages

- The H^{-a} norm accounts for both scale and intensity of segregation.
- decays even in the absence of diffusion.

Technical Definition I

Definition

The stirring field $u(x, t)$ is called **mixing** if for every periodic square-integrable function $g(x)$ on $[0, L]^d$,

$$\lim_{t \rightarrow \infty} \int_{[0, L]^d} g(x) \theta(x, t) dx = \langle g \rangle$$

Theorem

Suppose the spatially mean-zero function $\theta(x, t)$ is bounded uniformly in $L^2([0, L]^d)$ for all $t > 0$. Then

$$\lim_{t \rightarrow \infty} \int_{[0, L]^d} g(x) \theta(x, t) dx = 0 \quad \forall g \in L^2$$

$$\iff \lim_{t \rightarrow \infty} \|\theta(., t)\|_{H^{-a}} = 0$$

for any $a > 0$

Illustration III

Also, for $k = 0$, using the optimal velocity $u_p = -\frac{\Delta^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)}{\|\nabla^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)\|_{L^2}}$ where $\mathbb{P}(v) = v - \nabla\Delta^{-1}(\nabla \cdot v)$ using $\theta_0 = \sin x$

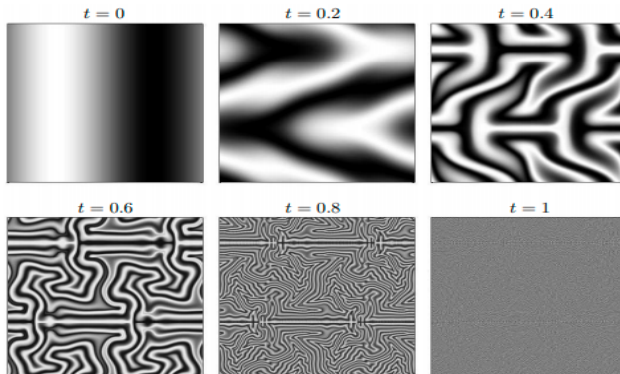


Figure: Evolution of the concentration field for the flow optimized in the case $a = 1$ as computed by Lin et al. Figure from Lin et al. [5]

PROBLEM STATEMENT

Advection-diffusion of a passive scalar field $\theta(x, t)$ by a smooth incompressible flow field $u(x, t)$ described by

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = k \Delta \theta, \quad \theta(x, 0) = \theta_0(x) \quad (0.5)$$

$$\nabla \cdot u = 0 \quad (0.6)$$

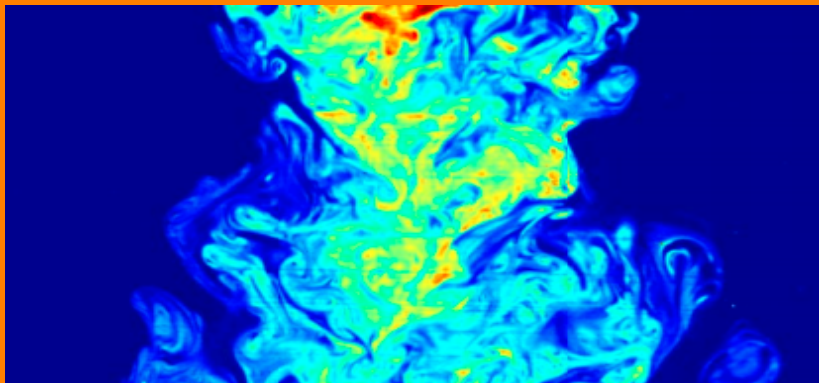
The stirring field u is periodic in the d -dimensional torus and L^∞ initial data θ_0 .

We consider the following cases

- ▶ $k = 0$
- ▶ $k \neq 0$
- ▶ $k \rightarrow 0$

- To study “how well” tracers can be mixed under a constraint in the advecting velocity field.

How well can tracers be mixed in the absence of diffusion?



MAIN PART I: MIXING WITHOUT DIFFUSION

For $k = 0$, we consider two constraints on u

- *Energy constraint* $\int_{[0,L]^d} |u(x, t)|^2 \, dx = U^2 L^d$
- *Enstrophy constraint* $\int_{[0,L]^d} |\nabla u(x, t)|^2 \, dx = \frac{L^d}{\tau^2}$

$$\begin{aligned} \frac{d}{dt} \|\theta\|_{H^{-1}}^2 &= \frac{d}{dt} \int |\nabla^{-1} \theta|^2 \, dx \\ &= \int \frac{\partial}{\partial t} |\nabla^{-1} \theta|^2 \, dx \\ &= \int 2 \nabla^{-1} \theta \frac{\partial}{\partial t} \nabla^{-1} \theta \, dx \\ &= \int 2 \nabla^{-1} \theta \nabla^{-1} \frac{\partial}{\partial t} \theta \, dx \end{aligned}$$

Using the incompressibility condition, we have

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 = -2 \int \theta u \cdot \nabla (\Delta^{-1} \theta) \, dx \quad (0.7)$$

MAIN PART I: MIXING WITHOUT DIFFUSION

Using Cauchy-Schwarz inequality we have

$$\begin{aligned}\frac{d}{dt} \|\theta\|_{H^{-1}}^2 &\geq -2 \|u\theta\|_{L^2} \|\nabla^{-1}\theta\|_{L^2} \\ &\geq -2 \|u\|_{L^2} \|\theta\|_{L^\infty} \|\nabla^{-1}\theta\|_{L^2}\end{aligned}$$

By the maximum principle, the L^∞ norm of θ is conserved i.e

$$\|\theta\|_{L^\infty} = \|\theta_0\|_{L^\infty}, \forall t > 0 \implies \frac{d}{dt} \|\theta\|_{H^{-1}}^2 \geq -2UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} \|\theta\|_{H^{-1}}$$

Divide by $2\|\theta\|_{H^{-1}}$ and integrating from $s = 0$ and $s = t$, we have

Energy constrained case: $\|\theta(., t)\|_{H^{-1}} \geq \|\theta(., 0)\|_{H^{-1}} - UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} t$
This lower bound clearly suggests that perfect mixing is possible in finite time.

MAIN PART I: MIXING WITHOUT DIFFUSION

For enstrophy constraint $\int_{[0,L]^d} |\nabla u(x, t)|^2 dx = \frac{L^d}{\tau^2}$, the lower bound on the mix-norm is given as follows:

$$\|\theta\|_{H^{-1}} \geq \|\theta_0\|_{H^{-1}} \left[1 - \frac{3C_3}{2\tau} \left(\frac{L\|\theta_0\|_{L^\infty}}{\|\theta_0\|_{H^{-1}}} \right)^{\frac{3}{2}} \times t \right]^{\frac{2}{3}} \quad (0.8)$$

provided the term in the square bracket is non-negative. *This also suggests that perfect mixing might be possible in finite time.*

MAIN PART I: MIXING WITHOUT DIFFUSION

The numerical simulations in [5], [2] show no perfect mixing in finite time for enstrophy bounded flow but a robust exponential decay of the mix norm. For $p = 2$, the Main Theorem 1 settles this discrepancy between simulation and analysis.

Main Theorem 1 (Gautam Iyer, Alexander Kiselev, Xiaoqian)

Let u be a smooth time independent incompressible periodic vector field on the d -dimensional torus, and let θ solve (0.4) with periodic boundary conditions and L_∞ initial data θ_0 .

For any $p > 1$ and $\lambda \in (0, 1)$ there exists a length scale $r_0 = r_0(\theta_0, \lambda)$, an explicit constant $\varepsilon_0 = \varepsilon_0(\lambda, d)$, and a constant $c = c(d, p)$ such that

$$\|\theta(t)\|_{H^{-1}} \geq \varepsilon_0 r_0^{\frac{d}{2}+1} \|\theta_0\|_{L^\infty} \exp\left(\frac{-c}{m(A_\lambda)^{\frac{1}{p}}} \int_0^t \|\nabla u(s)\|_{L^p} ds\right) \quad (0.9)$$

Here A_λ is the super level set $\{\theta_0 > \lambda \|\theta_0\|_{L^\infty}\}$

MAIN PART I: MIXING WITHOUT DIFFUSION

Enstrophy constrained Flow: Numerical Discussion

Here, we present numerical results illustrating how the exponential decay rate varies with the initial data.

We work on 1-periodic torus. Given a parameter b , consider the initial data θ_0 given as:

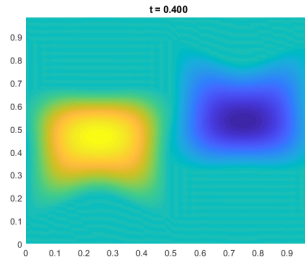
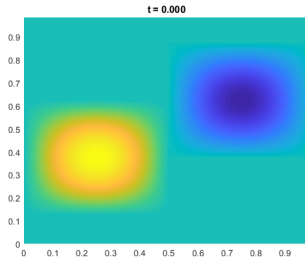
$$\theta_0(x, y) = \begin{cases} \sin\left(\frac{2\pi x}{b}\right) \sin\left(\frac{2\pi(y + \frac{b}{8})}{b}\right) & \text{for } 0 < x < \frac{b}{2} \text{ and } \frac{-b}{8} < y < \frac{b}{2} - \frac{b}{8} \\ \sin\left(\frac{2\pi x}{b}\right) \sin\left(\frac{2\pi(y - \frac{b}{8})}{b}\right) & \text{for } \frac{b}{2} < x < b \text{ and } \frac{b}{8} < y < \frac{b}{2} + \frac{b}{8} \\ 0 & \text{otherwise.} \end{cases}$$

We use the optimal velocity introduced by Zhi Lin, Jean-Luc, Charles Doering [5] using steepest descent method $u = -\frac{\Delta^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)}{\|\nabla^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)\|_{L^2}}$ where $\mathbb{P}(v) = v - \nabla\Delta^{-1}(\nabla \cdot v)$

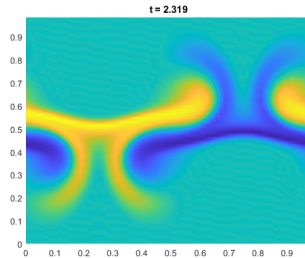
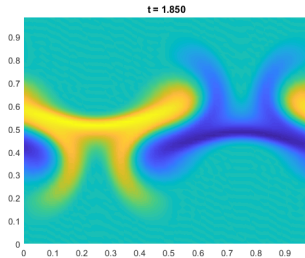
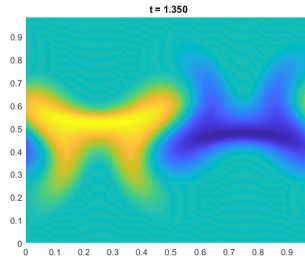
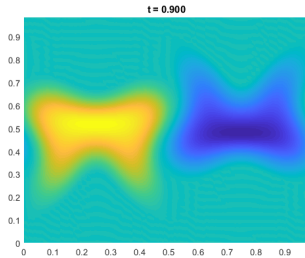
MAIN PART I: MIXING WITHOUT DIFFUSION

Using a pseudo-spectral method , a numerical simulation of (0.4) with the initial value obtained by varying b over the set $\{\frac{1}{4}, \frac{7}{12}, \dots, 1\}$.

The figures below show the solution plots for $b = 1$

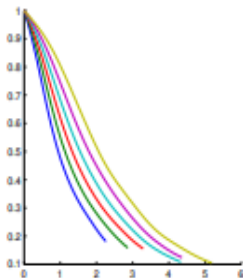


MAIN PART I: MIXING WITHOUT DIFFUSION

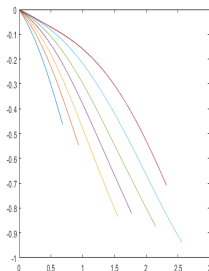


MAIN PART I: MIXING WITHOUT DIFFUSION

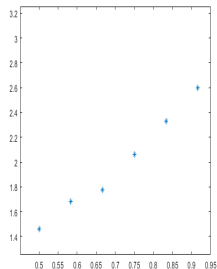
- Figure (a) below shows the exponential decay of the mix norm.
- Since $m(\text{supp}(\theta_0)) = O(b^2)$, Main theorem 1 predicts this graph as linear function of b .
- Fit a straight line to the log plots in Figure (b). Plot the negative reciprocal of the slope vs b in Figure (c).
- Figure (c) shows a linear relationship. This agrees with the analysis.



(a) $\|\theta(t)\|_{H^{-1}}$ vs t

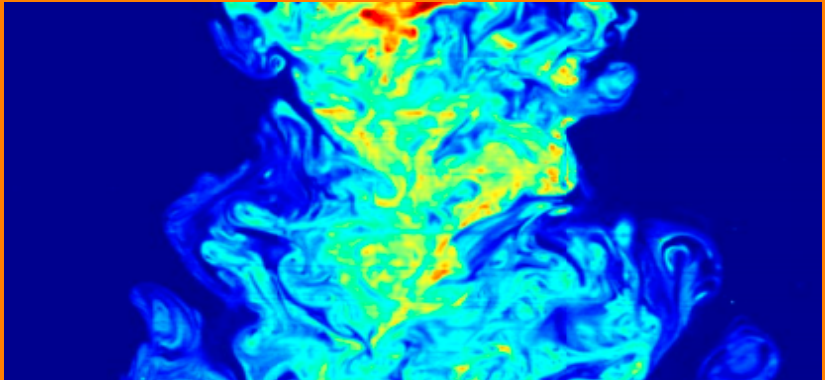


(b) $\ln \|\theta(t)\|_{H^{-1}}$ vs t



(c) Exponential decay rate.

Does the presence of diffusion aid mixing under constraints on the velocity field?



MAIN PART II: MIXING WITH DIFFUSION

$$\partial_t \theta + u \cdot \nabla \theta = k \Delta \theta \quad k \neq 0 \quad (0.10)$$

- **Enstrophy constrained** $\|\nabla u\|_{L^\infty} = 1$:

$$\|\nabla^{-1} \theta\|_{L^2} \geq \|\nabla^{-1} \theta_0\|_{L^2} \exp \left[-t - \frac{k}{2} \frac{\|\nabla \theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} (e^{2t} - 1) \right]$$

This suggests that perfect mixing in finite time is not possible for L^∞ enstrophy constrained flow.

- **Energy constrained** $\|u\|_{L^\infty} = 1$:

$$\|\nabla^{-1} \theta\|_{L^2} \geq \frac{\|\theta_0\|_{L^2}^2}{\|\nabla \theta_0\|_{L^2}^2} \exp \left[\frac{-t}{2k} - k^2 \frac{\|\nabla \theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} (e^{\frac{t}{k}} - 1) \right]$$

This also suggests that perfect mixing in finite time is not possible for L^∞ energy constrained flow

Anomalous Dissipation



Figure: Taken from the article 36 years ago, Mount St. Helens erupted, has helped scientists since. which appeared in The Oregonian, 05/2016

MAIN PART III: ANOMALOUS DISSIPATION

Consider

$$\partial_t \theta^\kappa + u \cdot \nabla \theta^\kappa = k \Delta \theta^\kappa \quad (0.11)$$

and multiply (0.11) by θ^κ , we have

$$\begin{aligned} \theta^\kappa \partial_t \theta^\kappa + \theta^\kappa u \cdot \nabla \theta^\kappa &= k \theta^\kappa \Delta \theta^\kappa \\ \frac{1}{2} \int \frac{\partial |\theta^\kappa(t)|^2}{\partial t} &= -k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \\ \frac{1}{2} |\theta^\kappa(t)|_{L^2}^2 &= \frac{1}{2} |\theta^\kappa(0)|_{L^2}^2 - k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \end{aligned} \quad (0.12)$$

- Eq(0.12) shows that the L^2 energy dissipation can be measured by $k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds$.
- there exists a situation where $k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \geq \chi > 0$ for some constant $\chi > 0$ **independent of κ** . This situation is called ***anomalous dissipation***.

MAIN PART III: ANOMALOUS DISSIPATION

Theodore D. Drivas, Tarek M. Elgindi, Gautam Iyer and In-Jee Jeong 2020 [1] provide the following main theorem of anomalous dissipation

Main Theorem 2 (Data dependent rate and velocity field)

Fix $T > 0$, $d \geq 2$, $\alpha \in [0, 1)$, and a mean-zero $\theta_0 \in H^2(\mathbb{T}^d)$. There exists a divergence-free velocity field

$$u \in C^\infty([0, T] \times \mathbb{T}^d) \cap L^1([0, T]; C^\alpha(\mathbb{T}^d)) \cap L^\infty([0, T] \times \mathbb{T}^d),$$

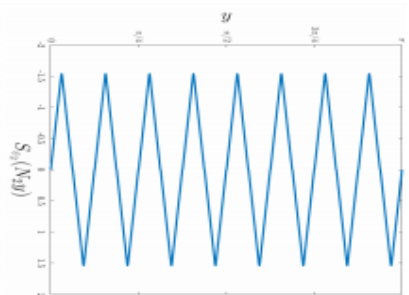
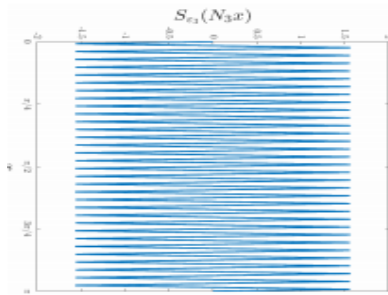
and $\chi_\alpha(\theta_0) > 0$ so that we have

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \geq \chi_\alpha(\theta_0) |\theta_0|_{L^2}^2. \quad (0.13)$$

MAIN PART III: ANOMALOUS DISSIPATION

Anomalous Dissipation relation with Mixing

- Anomalous dissipation is non-vanishing energy dissipation in the limit of zero viscosity. Turbulence is a super dissipator, capable of anomalous dissipation.
- Mixing requires all energy to be sent to high frequencies.
- Turbulent flows are exceptionally strong mixers.



CONCLUSION

- In the absence of diffusion, finite-time perfect mixing might
 - be possible by an energy constrained flows.
 - be impossible by an enstrophy constrained flows.
- In the presence of diffusion, finite-time perfect mixing might be impossible by an L^∞ energy and enstrophy constrained flows.
- Finally, we consider anomalous dissipation and its relation with mixing.

FUTURE WORK

In future work, we would like to:

- improve the lower bounds in case $k \neq 0$.
- consider the case of sources and sinks.
- consider the optimal control problem with finite-time optimisation to minimise the H^{-1} mix-norm at the end time rather than instantaneously attempting to minimise its decay rate. This might lead to flows that can produce even smaller length scales.

REFERENCES I

-  <https://www.altexsoft.com/blog/machine-learning-engineer/>
Anomalous Dissipation in Passive Scalar Transport
-  Gautam Iyer, Alexander Kiselev, Xiaoqian Xu (2014)
Lower bounds on the mix norm of passive scalars advected by incompressible enstrophy-constrained flows
-  C. J. Miles and C. R. Doering.
Diffusion-limited mixing by incompressible flows.
Nonlinearity, 31(5):2346, 2018.
-  www.math.cmu.edu/gautam/research/201208-mix-bounds
-  Z. Lin, J.-L. Thiffeault, and C. R. Doering.
Optimal stirring strategies for passive scalar mixing. *J. Fluid Mech.*, 2011
-  J.L. Thiffeault.
Using multiscale norms to quantify mixing and transport.
Nonlinearity, 2012.

REFERENCES II



P. V. Danckwerts,

The definition and measurement of some characteristics of mixtures, Appl Sci. Res (1952)



E. Lunasin, Z. Lin, A. Novikov, A. Mazzucato, and C. R. Doering.

Optimal mixing and optimal stirring for fixed energy, fixed power, or fixed palenstrophy flows. J. Math. Phys 2012



E. Gouillart, O. Dauchot, B. Dubrulle, S. Roux, and J.-L. Thiffeault.

Slow decay of concentration variance due to no-slip walls in chaotic mixing. Phys. Rev. 2008

Thank You